

Netflix Movies and TV Shows Data Analysis

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Introduction

This tiny project analyses Netflix catalogue data to understand the distribution of Movies and TV Shows, release-year patterns, ratings, countries, and genres. It demonstrates an end-to-end R workflow: data import, cleaning, statistical exploration, plots, interpretation, and a reproducible conclusion.

Research questions

1. Does the dataset contain more Movies or TV Shows?
2. Which release years, ratings, countries, and genres occur most frequently?
3. How has the number of titles released changed over time?

Data importing

Download a Netflix titles CSV file and save it as `netflix_titles.csv` in the same folder as this `.Rmd` file. This report expects the commonly used columns: `show_id`, `type`, `title`, `director`, `cast`, `country`, `date_added`, `release_year`, `rating`, `duration`, `listed_in`, and `description`.

```
netflix <- read.csv("netflix_titles.csv", stringsAsFactors = FALSE, na.strings = c("", "NA"))
head(netflix)
```

show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	descript
s1	Movie	The Silent City	Asha Mehta	Riya Shah	India	September 9, 2019	2018	TV-14	98 min	Dramas, International Movies	A fictiona drama.
s2	TV Show	Code Detectives	Rohan Patel	Jay Kumar	India, United States	January 12, 2020	2019	TV-PG	2 Seasons	TV Dramas, International TV Shows	A technol mystery series.
s3	Movie	Ocean Light	Maya Rao	Anika Bose	United States	March 15, 2021	2020	PG-13	112 min	Documentaries, International Movies	A document about oceans.
s4	Movie	Midnight Signal	Arjun Shah	Vikram Das	India	July 1, 2021	2021	TV-MA	105 min	Thrillers, International Movies	A suspense thriller.
s5	TV Show	Family Kitchen	Nina Thomas	Leena Roy	United Kingdom	August 20, 2022	2022	TV-G	1 Season	Reality TV, International TV Shows	A cooking competi
s6	Movie	Future Earth	Sam Lee	Kai Chen	South Korea, United States	November 11, 2022	2022	TV-14	120 min	Sci-Fi & Fantasy, Dramas	A science fiction stc

```
str(netflix)
```

```
## 'data.frame':   10 obs. of  12 variables:
## $ show_id      : chr  "s1" "s2" "s3" "s4" ...
## $ type         : chr  "Movie" "TV Show" "Movie" "Movie" ...
## $ title        : chr  "The Silent City" "Code Detectives" "Ocean Light" "Midnight Signal" ...
## $ director     : chr  "Asha Mehta" "Rohan Patel" "Maya Rao" "Arjun Shah" ...
## $ cast         : chr  "Riya Shah" "Jay Kumar" "Anika Bose" "Vikram Das" ...
## $ country      : chr  "India" "India, United States" "United States" "India" ...
## $ date_added   : chr  "September 9, 2019" "January 12, 2020" "March 15, 2021" "July 1, 2021" ...
## $ release_year : int   2018 2019 2020 2021 2022 2022 2023 2021 2024 2024
## $ rating       : chr  "TV-14" "TV-PG" "PG-13" "TV-MA" ...
## $ duration     : chr  "98 min" "2 Seasons" "112 min" "105 min" ...
## $ listed_in    : chr  "Dramas, International Movies" "TV Dramas, International TV Shows" "Documentaries, International Movies" "Thrillers, International Movies" ...
## $ description  : chr  "A fictional drama." "A technology mystery series." "A documentary about oceans." "A suspense thriller." ...
```

```
summary(netflix)
```

```
##      show_id      type      title      director      cast
## Length   :10 Length   :10 Length   :10 Length   :10 Length   :10
## N.unique :10 N.unique : 2 N.unique :10 N.unique :10 N.unique : 9
## N.blank  : 0 N.blank  : 0 N.blank  : 0 N.blank  : 0 N.blank  : 0
## Min.nchar: 2 Min.nchar: 5 Min.nchar:11 Min.nchar: 7 Min.nchar: 7
## Max.nchar: 3 Max.nchar: 7 Max.nchar:17 Max.nchar:11 Max.nchar:10
##
##      country      date_added release_year      rating      duration
## Length   :10 Length   :10 Min.    :2018 Length   :10 Length   :10
## N.unique : 7 N.unique :10 1st Qu.:2020 N.unique : 6 N.unique : 9
## N.blank  : 0 N.blank  : 0 Median  :2022 N.blank  : 0 N.blank  : 0
## Min.nchar: 5 Min.nchar:12 Mean    :2021 Min.nchar: 4 Min.nchar: 6
## Max.nchar:26 Max.nchar:17 3rd Qu.:2023 Max.nchar: 5 Max.nchar: 9
##
##              Max.    :2024
##
##      listed_in      description
## Length   :10 Length   :10
## N.unique :10 N.unique :10
## N.blank  : 0 N.blank  : 0
## Min.nchar:21 Min.nchar:15
## Max.nchar:35 Max.nchar:29
##
```

Preprocessing and cleaning

This section standardises text values, converts dates, removes duplicate titles, and removes records with invalid release years. Missing director, cast, country, rating, and duration values are replaced by `Unknown` so they can be counted rather than silently excluded.

```
required_columns <- c("show_id", "type", "title", "director", "cast", "country",
                     "date_added", "release_year", "rating", "duration", "listed_in", "description")
missing_columns <- setdiff(required_columns, names(netflix))
if (length(missing_columns) > 0) {
  stop("The CSV is missing: ", paste(missing_columns, collapse = ", "))
}

netflix_clean <- netflix %>%
  mutate(
    across(where(is.character), ~trimws(.x)),
    date_added = as.Date(date_added, format = "%B %d, %Y"),
    year_added = as.integer(format(date_added, "%Y")),
    release_year = as.integer(release_year),
    type = factor(type),
    rating = ifelse(is.na(rating) | rating == "", "Unknown", rating),
    country = ifelse(is.na(country) | country == "", "Unknown", country),
    listed_in = ifelse(is.na(listed_in) | listed_in == "", "Unknown", listed_in),
    director = ifelse(is.na(director) | director == "", "Unknown", director),
    duration = ifelse(is.na(duration) | duration == "", "Unknown", duration)
  ) %>%
  distinct(show_id, .keep_all = TRUE) %>%
  filter(!is.na(title), title != "", !is.na(release_year), release_year >= 1900, release_year <= 2100)

colSums(is.na(netflix_clean))
```

```
##      show_id      type      title      director      cast      country
##          0          0          0          0          0          0
##  date_added release_year      rating      duration      listed_in      description
##          0          0          0          0          0          0
##   year_added
##          0
```

```
kable(head(netflix_clean), caption = "First six cleaned records")
```

First six cleaned records

show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
s1	Movie	The Silent City	Asha Mehta	Riya Shah	India	2019-09-09	2018	TV-14	98 min	Dramas, International Movies	A fictional drama.
s2	TV Show	Code Detectives	Rohan Patel	Jay Kumar	India, United States	2020-01-12	2019	TV-PG	2 Seasons	TV Dramas, International TV Shows	A technological mystery series.
s3	Movie	Ocean Light	Maya Rao	Anika Bose	United States	2021-03-15	2020	PG-13	112 min	Documentaries, International Movies	A documentary about oceans.
s4	Movie	Midnight Signal	Arjun Shah	Vikram Das	India	2021-07-01	2021	TV-MA	105 min	Thrillers, International Movies	A suspense thriller.
s5	TV Show	Family Kitchen	Nina Thomas	Leena Roy	United Kingdom	2022-08-20	2022	TV-G	1 Season	Reality TV, International TV Shows	A cooking competition show.
s6	Movie	Future Earth	Sam Lee	Kai Chen	South Korea, United States	2022-11-11	2022	TV-14	120 min	Sci-Fi & Fantasy, Dramas	A science fiction story.

Statistical exploration

```
overview <- netflix_clean %>%
  summarise(
    Total_titles = n(),
    Movies = sum(type == "Movie"),
    TV_Shows = sum(type == "TV Show"),
    Earliest_release = min(release_year),
    Latest_release = max(release_year),
    Unique_ratings = n_distinct(rating),
    Unique_countries = n_distinct(country)
  )
kable(overview, caption = "Netflix catalogue overview")
```

Netflix catalogue overview

Total_titles	Movies	TV_Shows	Earliest_release	Latest_release	Unique_ratings	Unique_countries
10	6	4	2018	2024	6	7

```
type_count <- netflix_clean %>% count(type, sort = TRUE)
rating_count <- netflix_clean %>% count(rating, sort = TRUE) %>% slice_head(n = 10)
kable(type_count, caption = "Titles by content type")
```

Titles by content type

type	n
Movie	6
TV Show	4

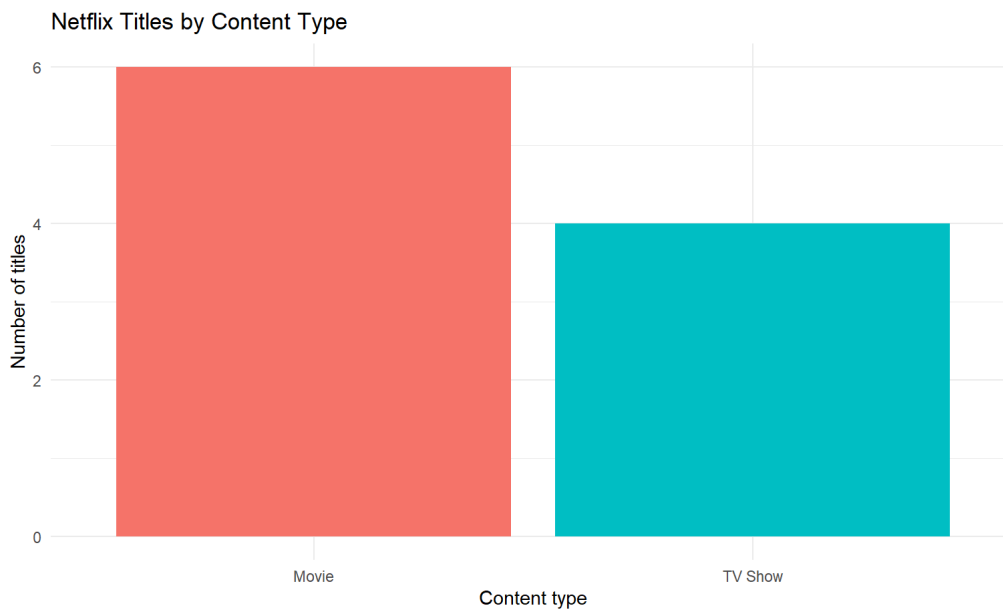
```
kable(rating_count, caption = "Ten most frequent ratings")
```

Ten most frequent ratings

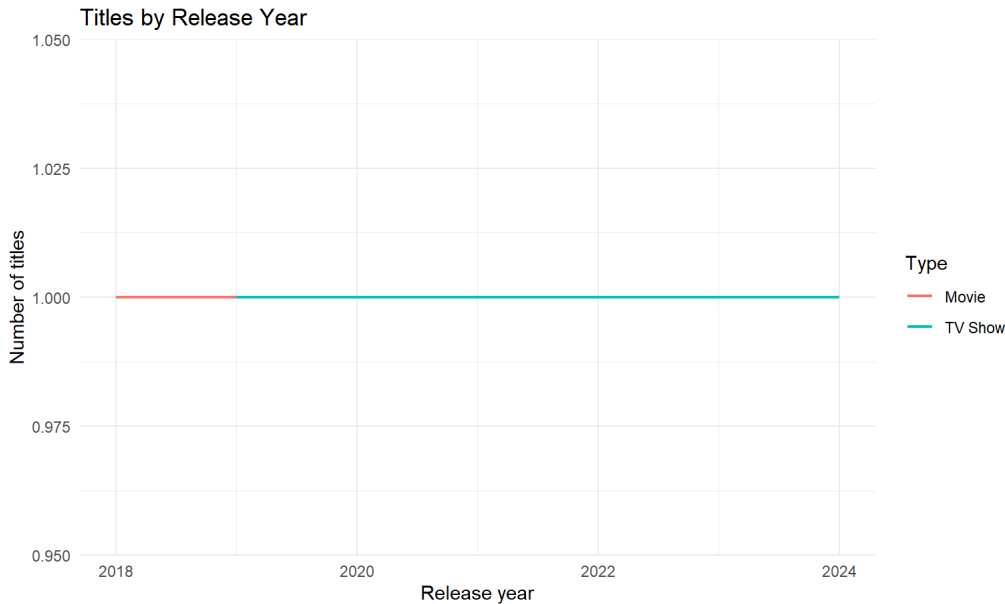
rating	n
TV-14	4
TV-PG	2
PG-13	1
TV-G	1
TV-MA	1
TV-Y7	1

Graphs and plots

```
ggplot(netflix_clean, aes(x = type, fill = type)) +  
  geom_bar() +  
  labs(title = "Netflix Titles by Content Type", x = "Content type", y = "Number of titles") +  
  theme_minimal() + theme(legend.position = "none")
```



```
netflix_clean %>%  
  count(release_year, type) %>%  
  ggplot(aes(x = release_year, y = n, colour = type)) +  
  geom_line(linewidth = 0.8) +  
  labs(title = "Titles by Release Year", x = "Release year", y = "Number of titles", colour = "Type") +  
  theme_minimal()
```



The line chart shows the catalogue composition by original release year. A fall in the final years may reflect incomplete data or titles not yet fully represented in the source file, rather than an actual production decline.

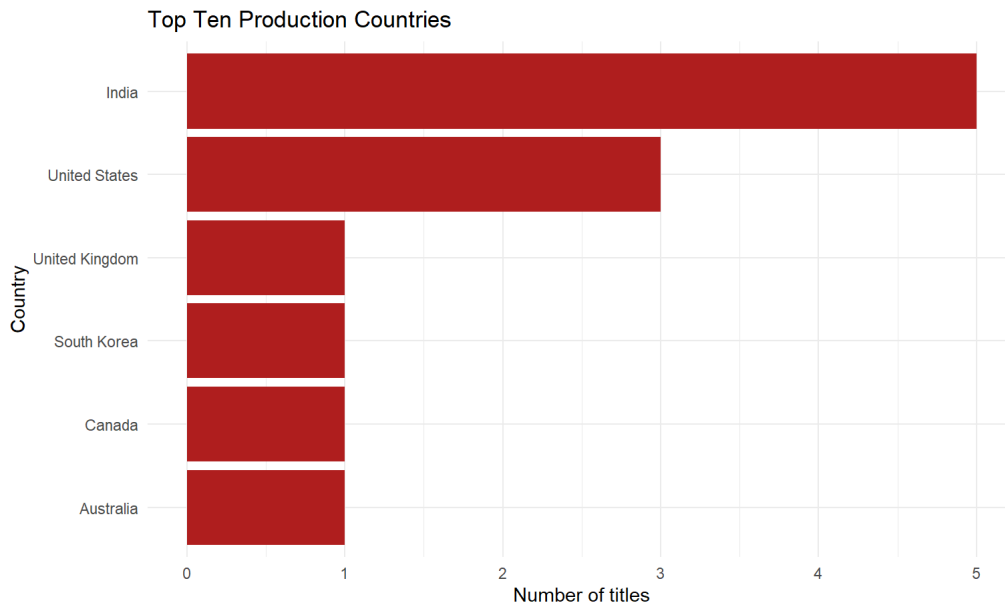
```
top_countries <- netflix_clean %>%
  filter(country != "Unknown") %>%
  mutate(country = str_split(country, "\\s*")) %>%
  tidyr::unnest(country) %>%
  count(country, sort = TRUE) %>%
  slice_head(n = 10)

kable(top_countries, caption = "Top ten production countries")
```

Top ten production countries

country	n
India	5
United States	3
Australia	1
Canada	1
South Korea	1
United Kingdom	1

```
ggplot(top_countries, aes(x = reorder(country, n), y = n)) +
  geom_col(fill = "firebrick") + coord_flip() +
  labs(title = "Top Ten Production Countries", x = "Country", y = "Number of titles") +
  theme_minimal()
```



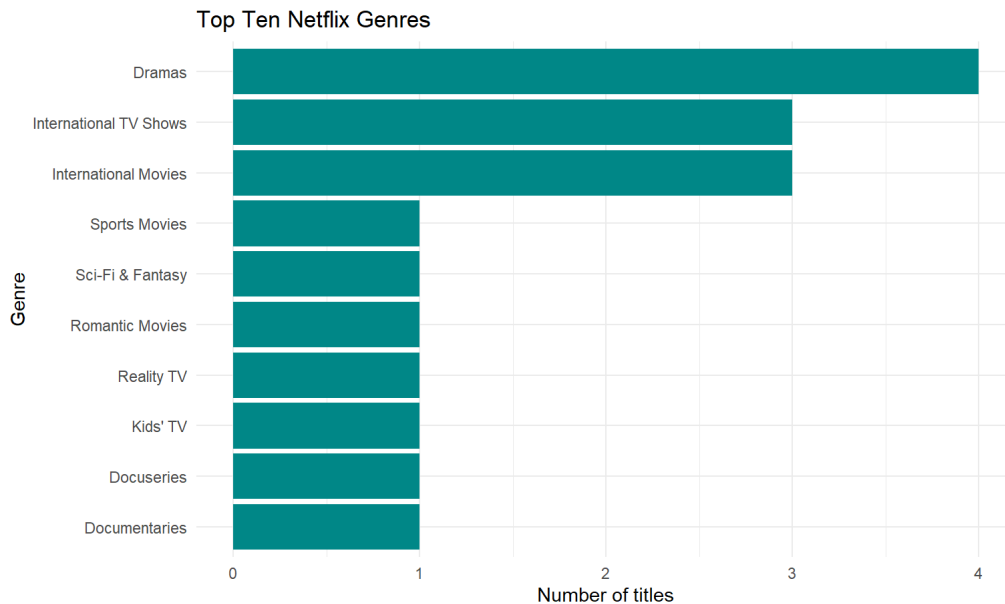
```
top_genres <- netflix_clean %>%
  filter(listed_in != "Unknown") %>%
  mutate(genre = str_split(listed_in, "\\s*")) %>%
  tidyr::unnest(genre) %>%
  count(genre, sort = TRUE) %>%
  slice_head(n = 10)

kable(top_genres, caption = "Top ten listed genres")
```

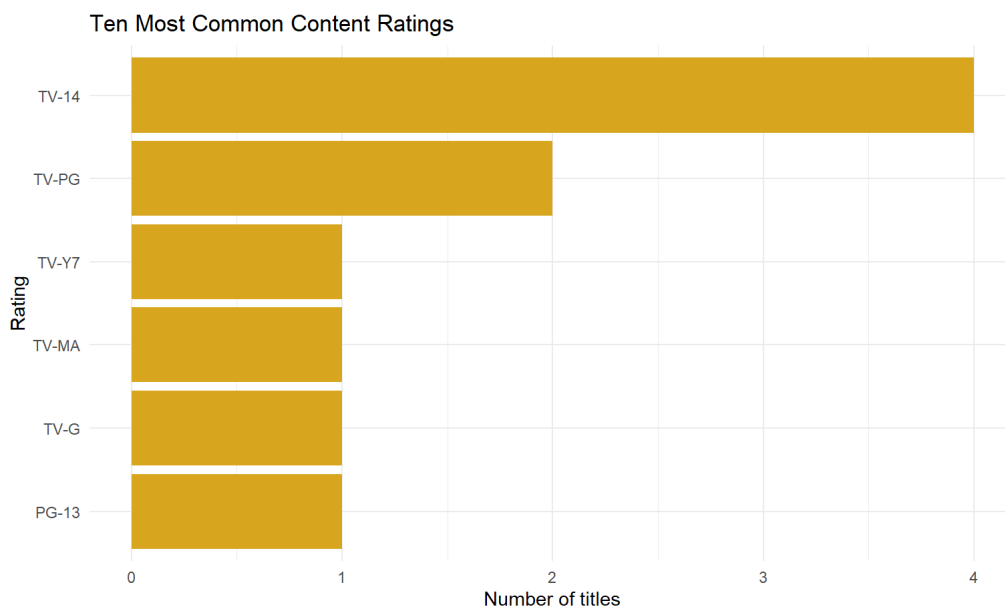
Top ten listed genres

genre	n
Dramas	4
International Movies	3
International TV Shows	3
Documentaries	1
Docuseries	1
Kids' TV	1
Reality TV	1
Romantic Movies	1
Sci-Fi & Fantasy	1
Sports Movies	1

```
ggplot(top_genres, aes(x = reorder(genre, n), y = n)) +
  geom_col(fill = "darkcyan") + coord_flip() +
  labs(title = "Top Ten Netflix Genres", x = "Genre", y = "Number of titles") +
  theme_minimal()
```



```
top_ratings <- netflix_clean %>% count(rating, sort = TRUE) %>% slice_head(n = 10)
ggplot(top_ratings, aes(x = reorder(rating, n), y = n)) +
  geom_col(fill = "goldenrod") + coord_flip() +
  labs(title = "Ten Most Common Content Ratings", x = "Rating", y = "Number of titles") +
  theme_minimal()
```



Interpretation of results

- The content-type chart directly answers whether Movies or TV Shows dominate the selected Netflix catalogue.
- Country and genre fields can contain more than one value per title; this report separates comma-delimited values before counting, so a co-produced title can contribute to multiple countries or genres.
- The release-year plot describes titles available in the dataset, not Netflix subscriber demand, viewing hours, or revenue.
- Missing metadata is labelled `Unknown` ; therefore, the analysis makes data quality visible instead of treating missing information as a real category.

Findings

- The overview table reports the total number of titles and the split between Movies and TV Shows.
- The release-year chart highlights periods with more titles represented in the catalogue.
- The country and genre charts identify the most frequently represented production locations and categories.
- Results depend on the date, country, and source of the downloaded Netflix dataset; the catalogue changes over time.

Conclusion

This project successfully applies data import, preprocessing, descriptive analysis, and data visualisation to a Netflix catalogue dataset. The analysis provides a useful overview of content type, release periods, ratings, production countries, and genres. A future extension could analyse the relationship between content rating and genre, compare Movies with TV Shows by country, or join the data with IMDb/TMDb ratings where permitted.

Reproducibility checklist

- Put `netflix_titles.csv` and this `.Rmd` file in the same folder.
- Install packages with `install.packages(c("rmarkdown", "knitr", "dplyr", "ggplot2", "stringr", "tidyr"))`.
- Open the `.Rmd` file in RStudio and select **Knit > Knit to HTML**.
- Verify that the HTML report is created and every table and plot is visible before submission.